



SYMBIOSIS INTERNATIONAL (DEEMED UNIVERSITY)

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Seat No.							
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Institute: (0203)SYMBIOSIS CENTRE FOR MANAGEMENT AND HUMAN RESOURCE DEVELOPMENT,
PUNE
Programme: (020341) MASTER OF BUSINESS ADMINISTRATION

Batch: 2024-26
Semester: I
Course: Research Methodology
Course Code: 0203410109

Date: 12/03/2025

Maximum Marks: 40

Day: Wednesday

Time: 14:00 - 15:30

NOTE : DO NOT WRITE ANYTHING ON THE QUESTION PAPER

Instructions:

- All questions are compulsory.

Q.1 A researcher is planning a psychological intervention study, but before he proceeds, he wants to characterize his participants' depression levels. He tests each participant on a particular depression index, where anyone who achieves a score of 4.0 is deemed to have 'normal' levels of depression. Lower scores indicate less depression, and higher scores indicate greater depression. He has recruited 40 participants to take part in the study.

Answer the following questions:

(10 Marks)

- Formulate hypothesis (CO2)
- Write assumptions of the one-sample t-test (CO3)
- Write the results and a detailed interpretation of each table given below. (CO1; CO4)

Table 1.1: Tests of Normality						
	Kolmogorov-Smirnov ^a			Shapiro-Wilk		
	Statistic	df	Sig.	Statistic	df	Sig.
Score	.273	40	.000	.803	40	.000
a. Lilliefors Significance Correction						

Table 1.2: One-Sample Statistics				
	N	Mean	Std. Deviation	Std. Error Mean
Score	40	2.8740	.52901	.08364

Table 1.3: One-Sample Test						
	Test Value = 4					
	t	df	Sig. (2-tailed)	Mean Difference	95% Confidence Interval of the Difference	
					Lower	Upper
Score	-13.462	39	.000	-1.12600	-1.2952	-.9568

- Q.2** A retailer wants to measure the effect of promotions on the sales of a particular brand of detergent. To check out the effect, he conducts an experiment in his retail store. For weeks 1, 3, and 6, he returns a sales promotion. Weekly sales for all six weeks of the detergent are tabulated. He then repeats the whole experiment two times for twelve more weeks—six weeks with a promotion and six weeks without a promotion.

Answer the following questions:

(15 Marks)

- Formulate hypothesis (CO2)
- Write assumptions of the independent sample t-test (CO3)
- Write the results and a detailed interpretation of each table given below. (CO1; CO4)

Table 2.1: Tests of Normality							
	Promotions	Kolmogorov-Smirnov ^a			Shapiro-Wilk		
		Statistic	df	Sig.	Statistic	df	Sig.
Sales	On	.342	9	.003	.678	9	.001
	Off	.128	9	.200*	.962	9	.822
*. This is a lower bound of the true significance.							
a. Lilliefors Significance Correction							

Table 2.2: Group Statistics					
	Promotions	N	Mean	Std. Deviation	Std. Error Mean
Sales	On	9	78.2222	27.57162	9.19054
	Off	9	55.6667	12.94218	4.31406

Table 2.3: Independent Samples Test

		Levene's Test for Equality of Variances		t-test for Equality of Means						
		F	Sig.	t	df	Sig. (2-tailed)	Mean Difference	Std. Error Difference	95% Confidence Interval of the Difference	
									Lower	Upper
Sal es	Equal variances assumed	.735	.404	2.22 2	16	.041	22.5555 6	10.1526 9	1.03281	44.0783 0
	Equal variances not assumed			2.22 2	11.3 62	.047	22.5555 6	10.1526 9	.29623	44.8148 8

Q.3 A psychologist wishes to assess the efficacy of a new cognitive-behavioural treatment for generalised anxiety. Before beginning the treatment with a new client, she asks him/her to complete the *Beck Anxiety Inventory (BAI)*. The pre-treatment *BAI* scores for 12 clients are listed in the first column of the given table. Scores on this measure can range from 0 through to 63.

At the end of the eight-week treatment program, the psychologist asked each client to again complete the *BAI*. The post-treatment *BAI* scores for the 10 clients who completed the full eight-week program are listed in the second column of the given table. Post-treatment *BAI* data are not available for two clients who dropped out of the program after two and five weeks, respectively.

Answer the following questions:

(15 Marks)

- Formulate hypothesis (CO2)
- Write assumptions of the paired sample t-test (CO3)
- Write the results and a detailed interpretation of each table given below. (CO1; CO4)

Table 3.1: Tests of Normality

	Kolmogorov-Smirnov ^a			Shapiro-Wilk		
	Statistic	df	Sig.	Statistic	df	Sig.
Pre-Treatment BAI	.137	10	.200*	.957	10	.750
Post-Treatment BAI	.139	10	.200*	.983	10	.980
*. This is a lower bound of the true significance.						
a. Lilliefors Significance Correction						

Table 3.2: Paired Samples Statistics					
		Mean	N	Std. Deviation	Std. Error Mean
Pair 1	Pre-Treatment BAI	44.30	10	7.499	2.371
	Post-Treatment BAI	41.80	10	6.779	2.144

Table 3.3: Paired Samples Test									
		Paired Differences					t	df	Sig. (2-tailed)
		Mean	Std. Deviation	Std. Error Mean	95% Confidence Interval of the Difference				
					Lower	Upper			
Pair 1	Pre-Treatment BAI - Post-Treatment BAI	2.500	3.567	1.128	-.052	5.052	2.216	9	.054